

How to measure wheat seed shape

If the number of images is large, you may exhaust Google Colab's free quota. In that case, either purchase Compute units or download the script to your local environment and run it in Jupyter Lab. With 100 Compute units, you can analyze thousands to tens of thousands of images; upload speed will be the rate-limiting factor.

- ① Sample data
- ② Print the template sheet and measure scale
- ③ Take picture
- ④ Run the script on google drive
- ④ data check and analyze

① Sample data

Access to

https://github.com/sou-06/Wheat_Seed_Phenotype/tree/main?tab=readme-ov-file

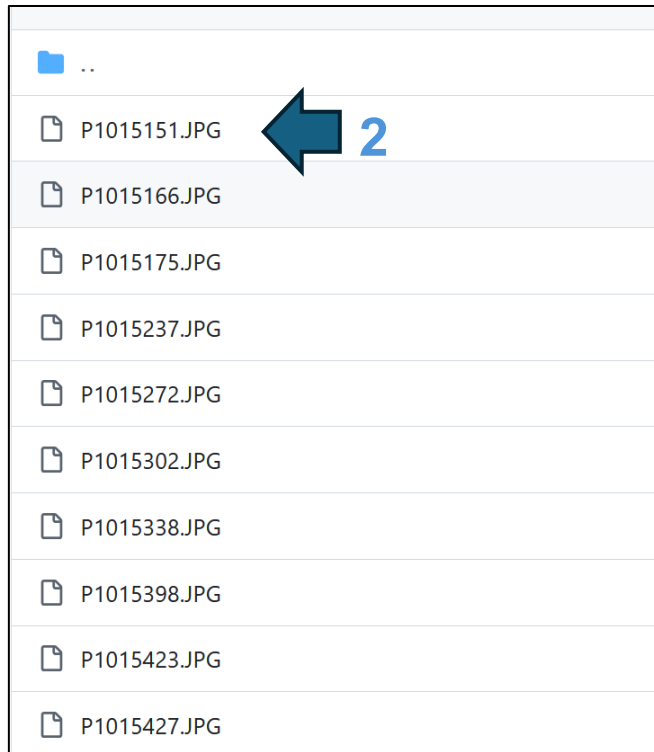
 **Wheat_Seed_Phenotype** PublicPinWatch

main 1 Branch 0 TagsGo to fileAdd fileCode

 **sou-06** Add files via upload daef975 · 32 minutes ago 11 Commits

 sampledata 	Add files via upload	32 minutes ago
 sampledata README.md	Update README.md	5 months ago
 Wheat_Seed.ipynb	Created using Colab	9 hours ago
 wheat_model.pt	Add files via upload	5 months ago
 wheat_model2.pt	Add files via upload	5 months ago

① - 1. Download sample data.



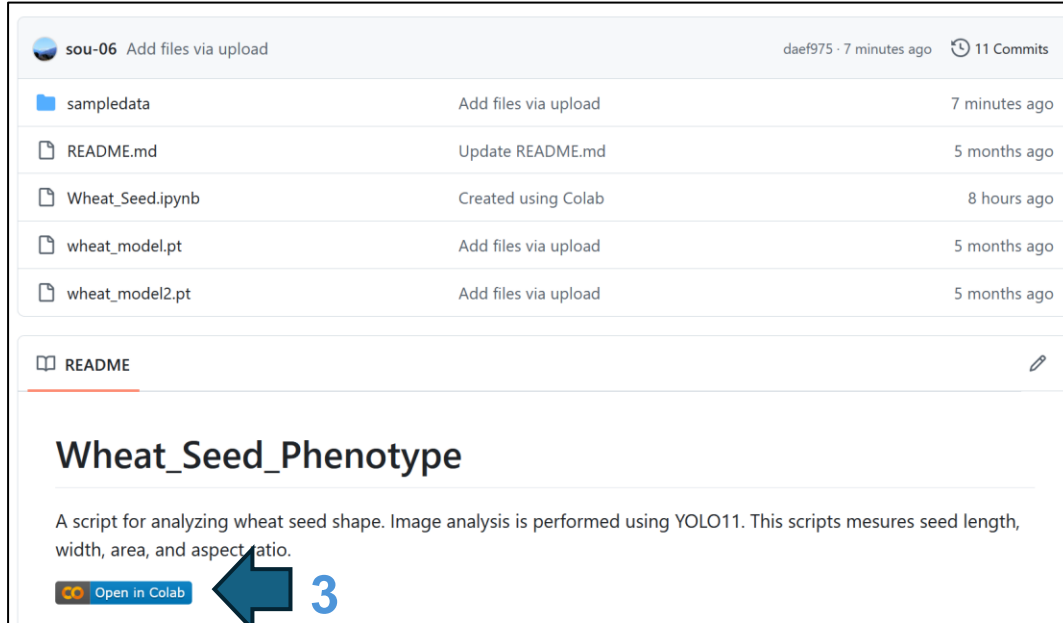
Repeating similar operations to download several different images (3 – 10 pictures).

① - 2. Upload sample data

Access to

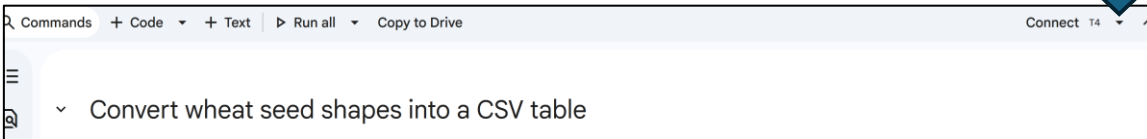
https://github.com/sou-06/Wheat_Seed_Phenotype/tree/main?tab=readme-ov-file

(2) Open google colab

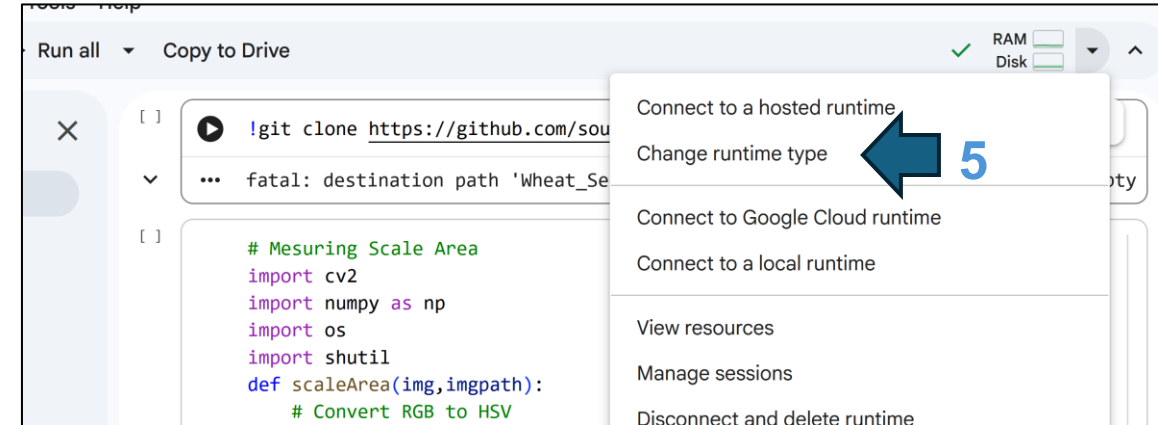


The screenshot shows a Google Colab notebook interface. At the top, there's a header with the user 'sou-06' and a file list. The file list includes 'sampledata', 'README.md', 'Wheat_Seed.ipynb', 'wheat_model.pt', and 'wheat_model2.pt'. Below the file list is a 'README' section titled 'Wheat_Seed_Phenotype'. The README text describes a script for analyzing wheat seed shape using YOLO11. A blue arrow labeled '3' points to the 'Open in Colab' button at the bottom left of the README section.

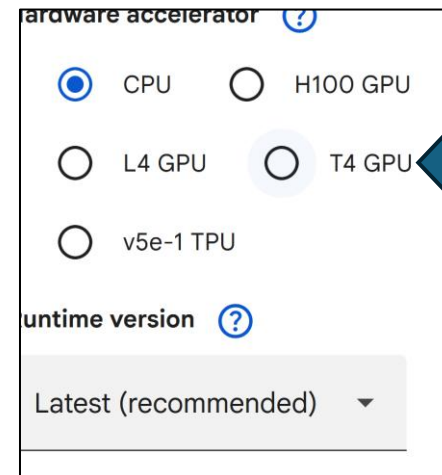
4



The screenshot shows the Google Colab interface with the 'Run all' button highlighted. A blue arrow labeled '4' points to the 'Run all' button. Below the 'Run all' button, there's a code cell with the text 'Convert wheat seed shapes into a CSV table'.



The screenshot shows a Google Colab interface with a runtime error. The error message is 'fatal: destination path 'Wheat_Se' is already under control'. A dropdown menu is open, showing options: 'Connect to a hosted runtime', 'Change runtime type', 'Connect to Google Cloud runtime', 'Connect to a local runtime', 'View resources', 'Manage sessions', and 'Disconnect and delete runtime'. A blue arrow labeled '5' points to the 'Change runtime type' option.

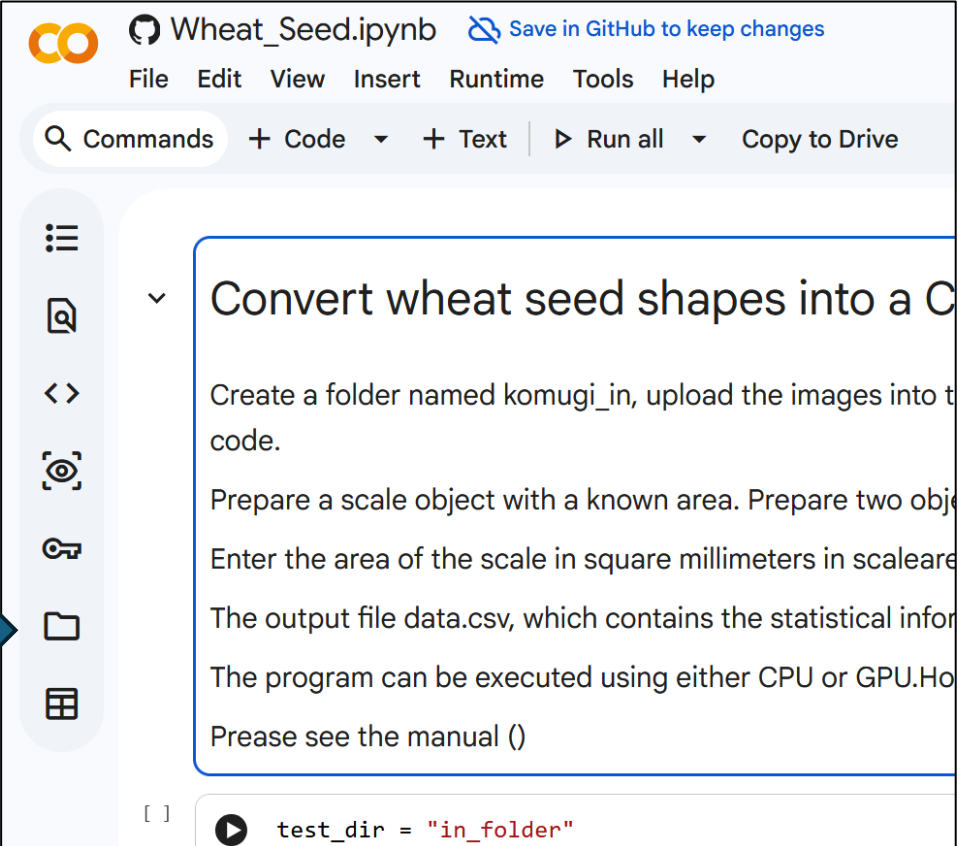


The screenshot shows the Google Colab hardware accelerator selection menu. The options are: CPU, H100 GPU, L4 GPU, T4 GPU, and v5e-1 TPU. The 'T4 GPU' option is selected. A blue arrow labeled '6' points to the 'T4 GPU' option. Below the hardware accelerator selection, there's a 'runtime version' dropdown menu with the option 'Latest (recommended)'.

6. Choose T4 GPU
->Save

① - 2. Upload sample data

7



Wheat_Seed.ipynb Save in GitHub to keep changes

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Copy to Drive

Convert wheat seed shapes into a C

Create a folder named komugi_in, upload the images into t
code.

Prepare a scale object with a known area. Prepare two obje

Enter the area of the scale in square millimeters in scaleare

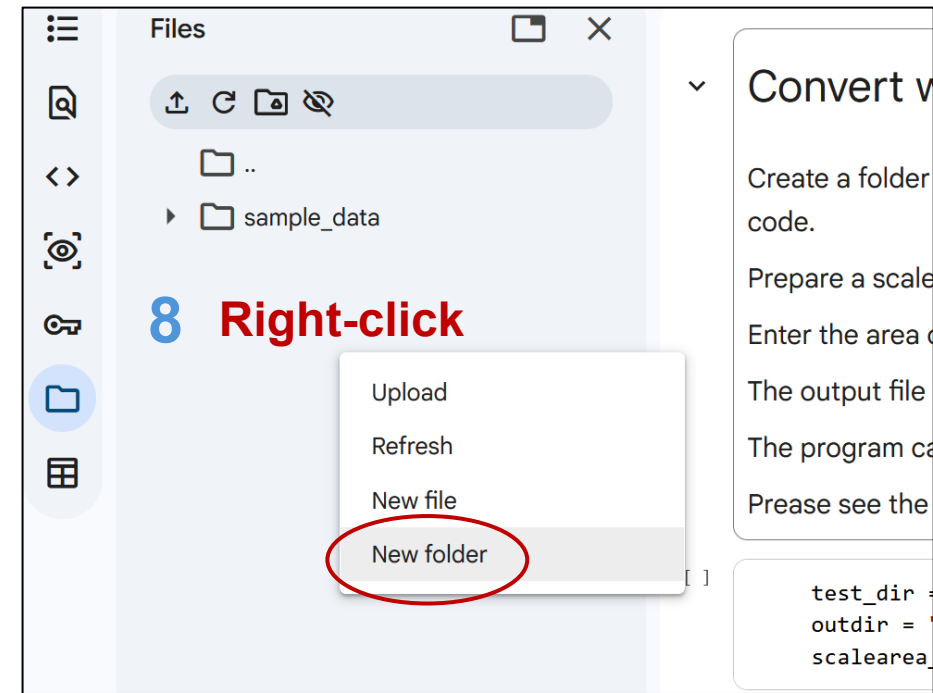
The output file data.csv, which contains the statistical infor

The program can be executed using either CPU or GPU.Ho

Prease see the manual ()

```
test_dir = "in_folder"
```

8 Right-click



Files

Upload Refresh New file New folder

Convert v

Create a folder
code.

Prepare a scale

Enter the area c

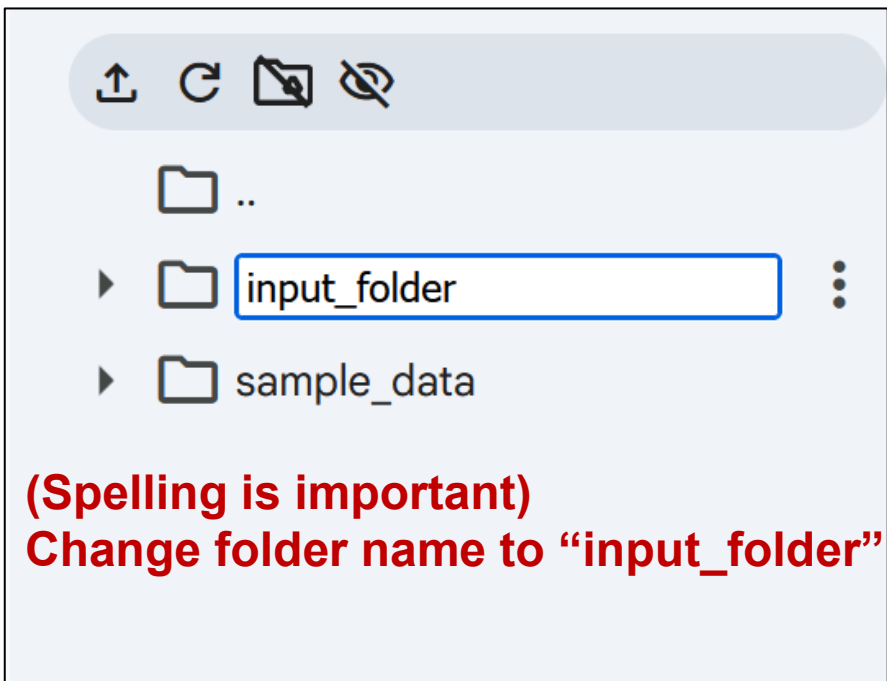
The output file

The program ca

Prease see the

```
test_dir =  
outdir =  
scalearea
```

9 (Spelling is important)
Change folder name to "input_folder"

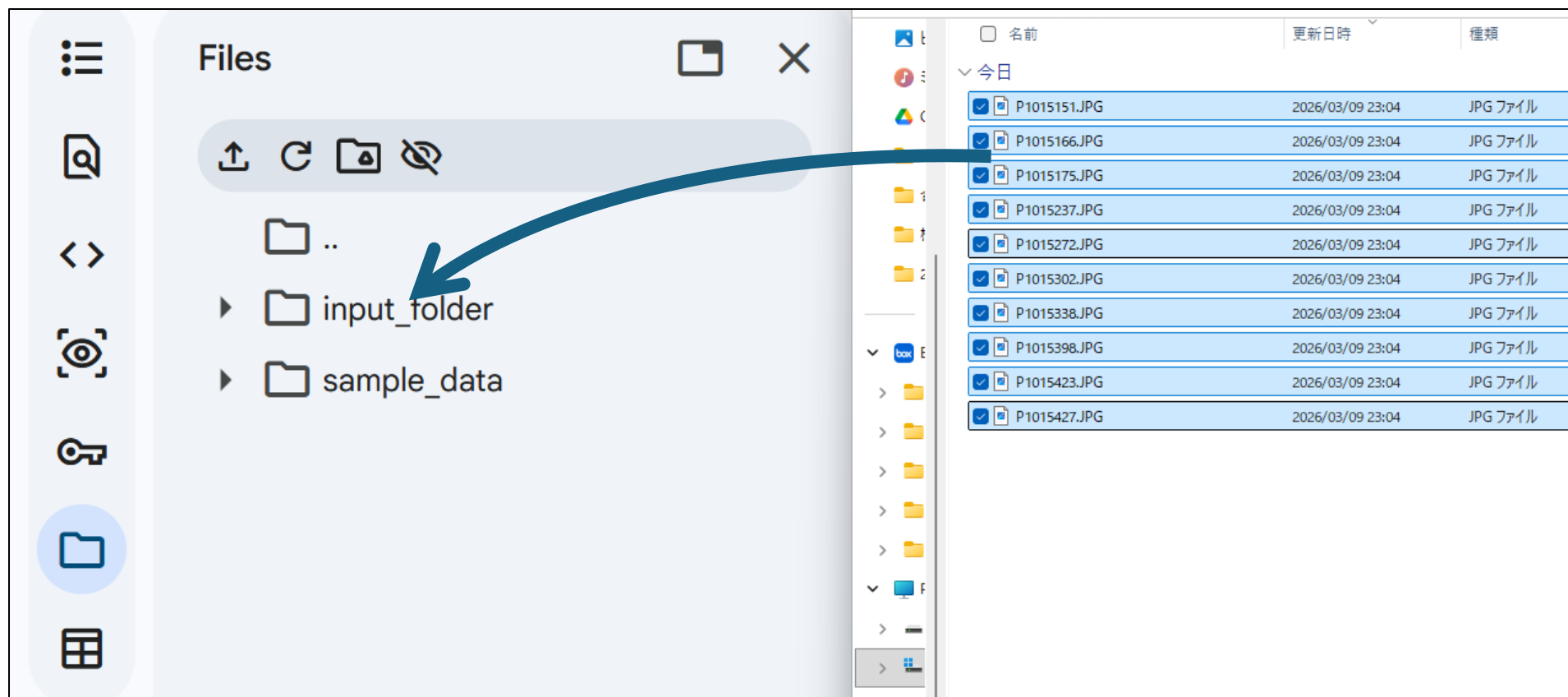


input_folder

sample_data

① - 2. Upload sample data

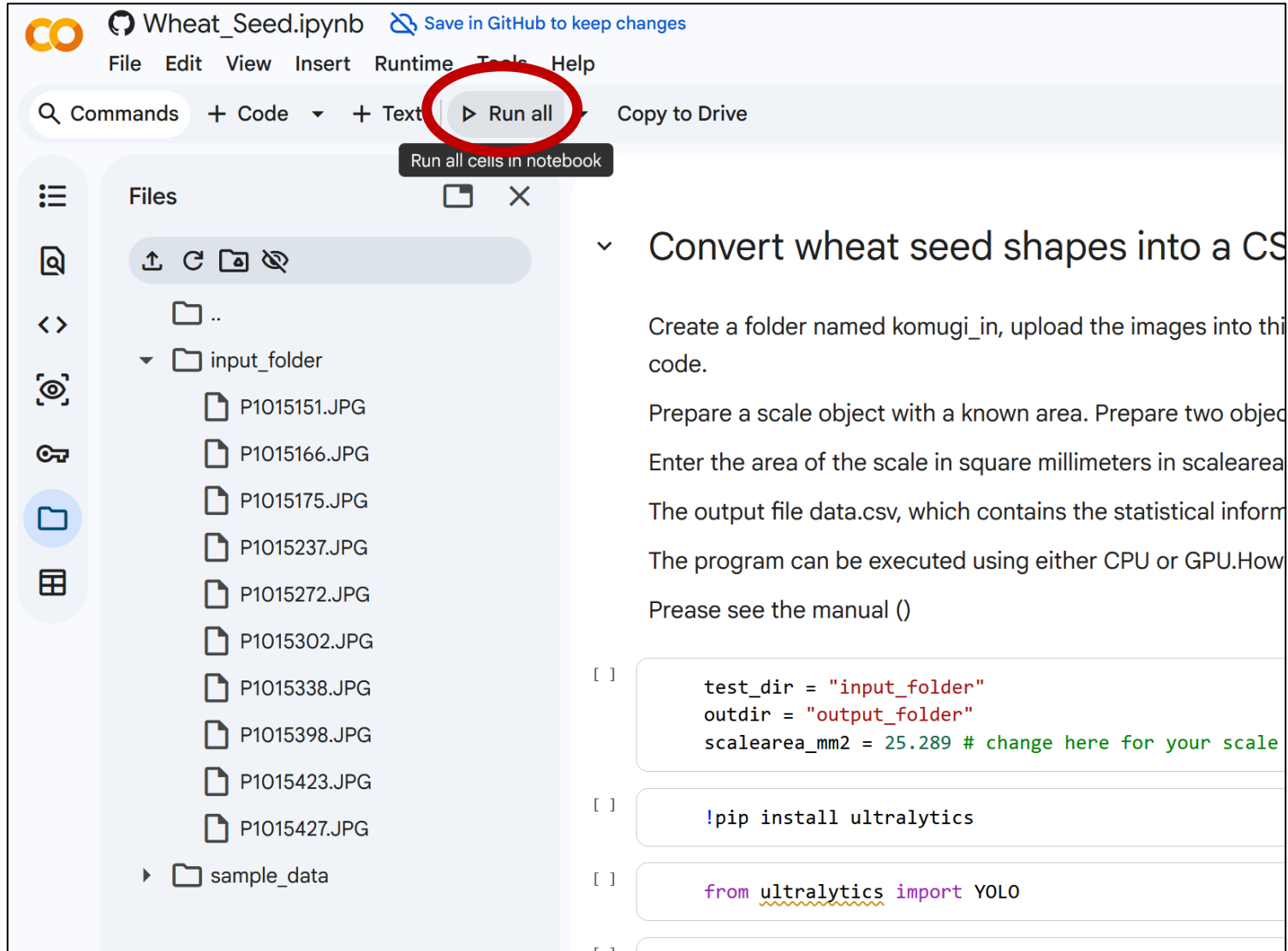
10. Drag and drop pictures into “input_folder”



① - 3. Run sample data

11. ▶ Run all

Please wait few minutes.



The screenshot shows a Jupyter Notebook interface for a file named 'Wheat_Seed.ipynb'. The top bar includes a search bar, a 'Commands' dropdown, and a 'Run all' button, which is circled in red. A tooltip for the 'Run all' button reads 'Run all cells in notebook'. The left sidebar displays a file explorer with a folder named 'input_folder' containing several JPEG files and a 'sample_data' folder. The main area shows the notebook content, which includes a title 'Convert wheat seed shapes into a CS', a description of the task, and code cells. The first code cell contains the following Python code:

```
[ ] test_dir = "input_folder"
    outdir = "output_folder"
    scalearea_mm2 = 25.289 # change here for your scale
```

The second code cell contains the command:

```
[ ] !pip install ultralytics
```

The third code cell contains the import statement:

```
[ ] from ultralytics import YOLO
```

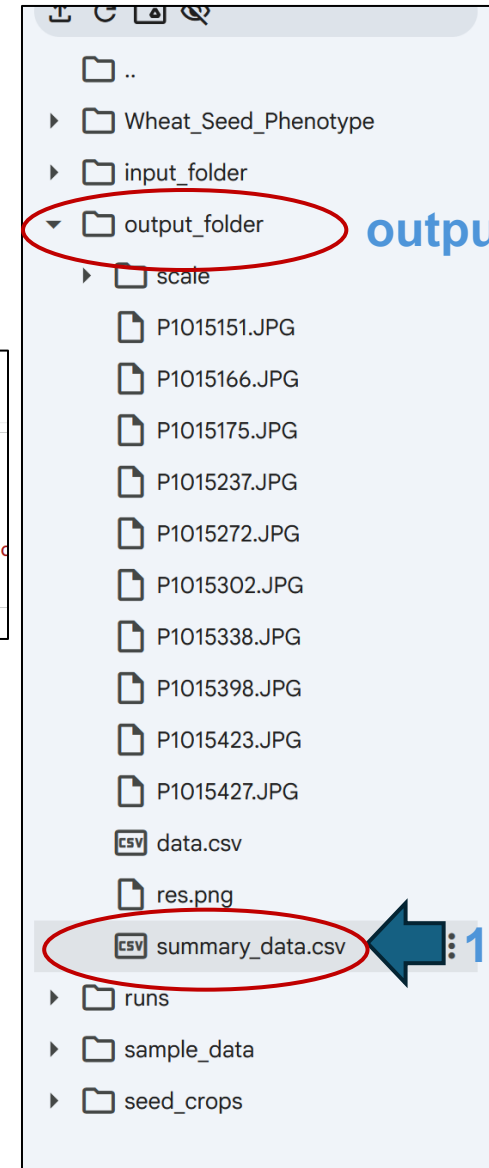
① - 4. Check result

12. Scroll down and wait for the cell for “Making summarized data” to complete.

```
✓ [8] 0s Making summarized data
import pandas as pd

df = pd.read_csv(outdir + "/data.csv")
result = df.groupby("File_Name")[["length_mm", "width_mm", "Area_mm2"]].agg(["mean", "std", "median", "count"])
result.to_csv(outdir + "/summary_data.csv")
```

Output_folder > **summary_data.csv**



output_folder

13. Right-click and download

① - 4. Check result

	A	B	C	D	E	F	G	H	I	J	K	L	M	
		length mm	length mm	length mm	length mm	width mm	width mm	width mm	width mm	Area mm2	Area mm2	Area mm2	Area mm2	
		mean	std	median	count	mean	std	median	count	mean	std	median	count	
File_Name														
P1015151		6.291882	0.537137	6.344463	90	3.071284	0.253556	3.051417	90	15.21821	2.126726	15.30431	90	
P1015166		6.545488	0.480854	6.595992	165	3.230528	0.289725	3.260713	165	16.66667	2.356914	16.8883	165	
P1015175		5.974785	0.440631	5.980284	89	3.044792	0.302523	3.054924	89	14.36147	2.226711	14.52029	89	
P1015237		5.780698	0.371865	5.804195	93	3.142547	0.290471	3.139018	93	14.32659	2.066813	14.47962	93	
P1015272		5.888534	0.475832	5.875616	57	3.113154	0.34306	3.142855	57	14.48763	2.491514	14.50333	57	
P1015302		5.961168	0.414797	5.936583	100	3.332691	0.320769	3.317596	100	15.65203	2.198473	15.55513	100	
P1015338		5.89249	0.439844	5.89982	126	2.799051	0.386438	2.793023	126	13.02064	2.394226	13.08756	126	
P1015398		6.269714	0.487978	6.331117	63	2.804117	0.408462	2.789094	63	13.87419	2.608743	14.25247	63	
P1015423		6.088394	0.468615	6.14058	64	2.817258	0.33201	2.718471	64	13.53726	2.303346	13.40427	64	
P1015427		6.247933	0.392415	6.316151	87	3.128766	0.353329	3.143161	87	15.42752	2.436478	15.86754	87	

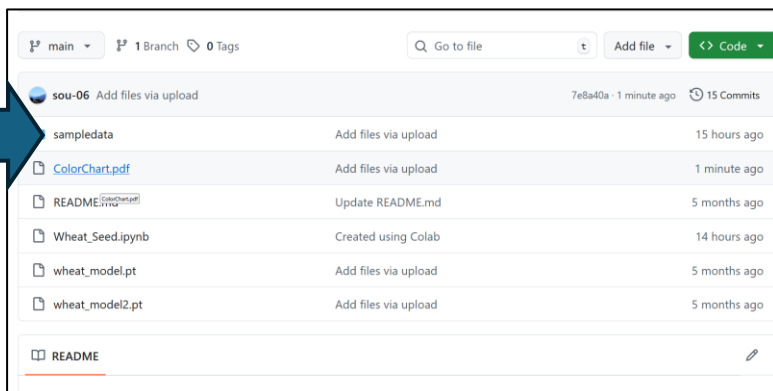
**For each image, lengths and widths are calculated in millimeters.
Count is the number of seeds detected**

② Print the template sheet and measure scale

② Print the template sheet and measure scale

The size of the markers varies depending on the printer, so the marker size should be set each time printing. **For comparative data, use the same paper or consecutively printed sheets and apply the same scale values. When using the same paper, please use the same value without remeasuring.**

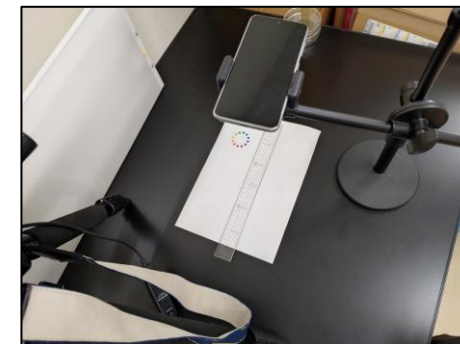
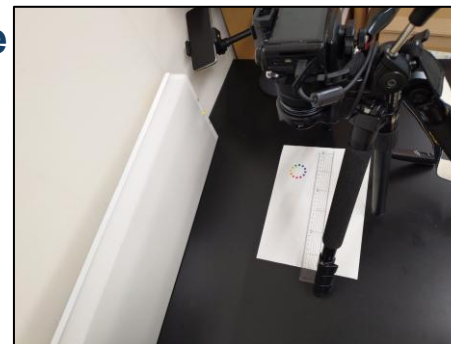
1



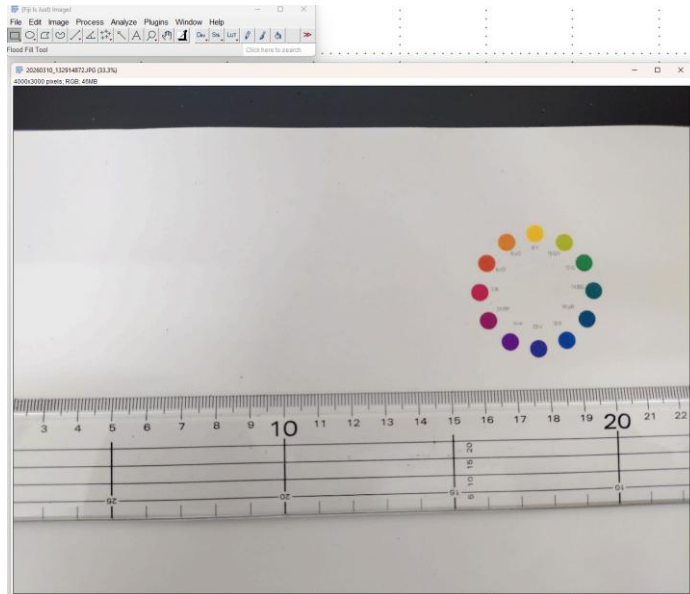
2. Download and printout



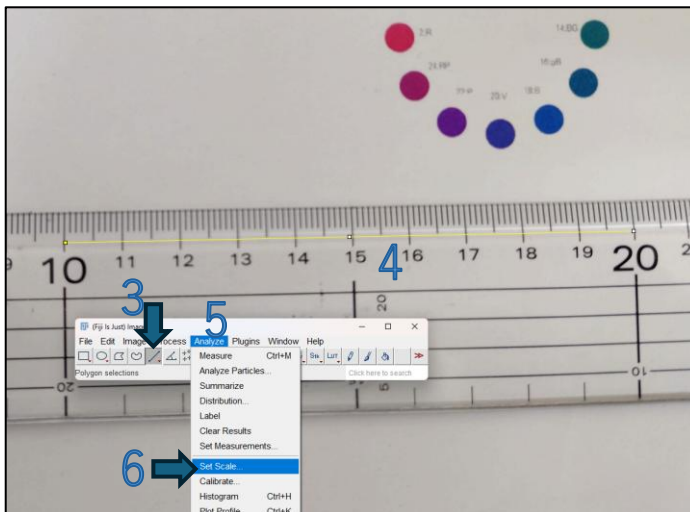
3. Take picture with measure.
Make sure the camera is level.



② Print the template sheet and measure scale



2. Open with ImageJ. (Fiji)



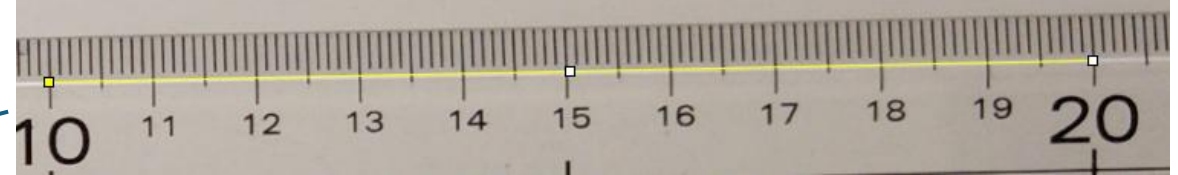
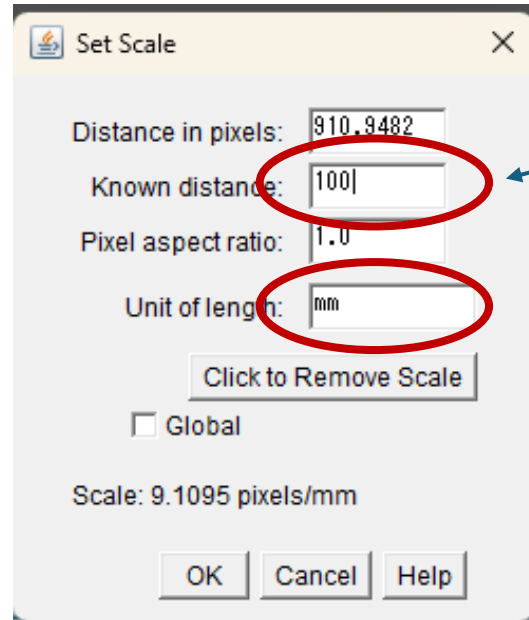
3. Straight

4. Draw a line along the ruler's markings

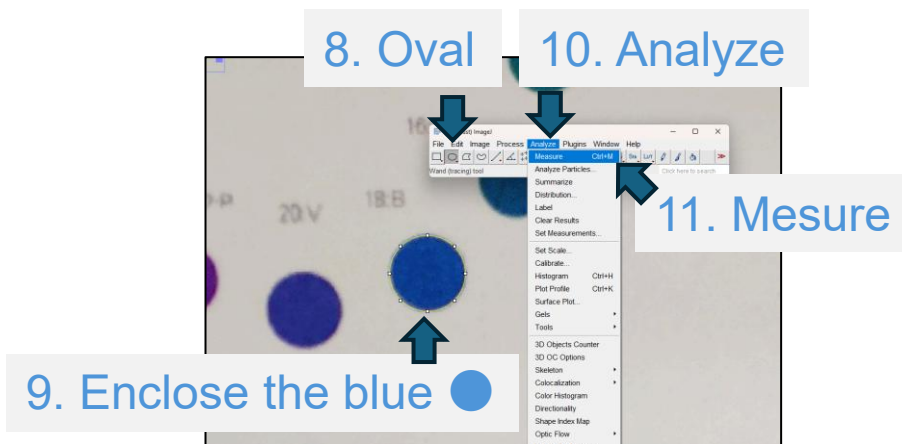
5. Analyze

6. Set scale

② Print the template sheet and measure scale



7. Set scale length with **mm**. -> OK.



Results										
File	Edit	Font	Results							
	Label	Area	Mean	Min	Max	XM	YM	Perim.	BX	
1	20260310_155425241.JPG	23.812	70.091	32	173	335.749	116.208	17.244	333.06	

12. Make a note of this value

③ Take picture

Make sure the camera is level.

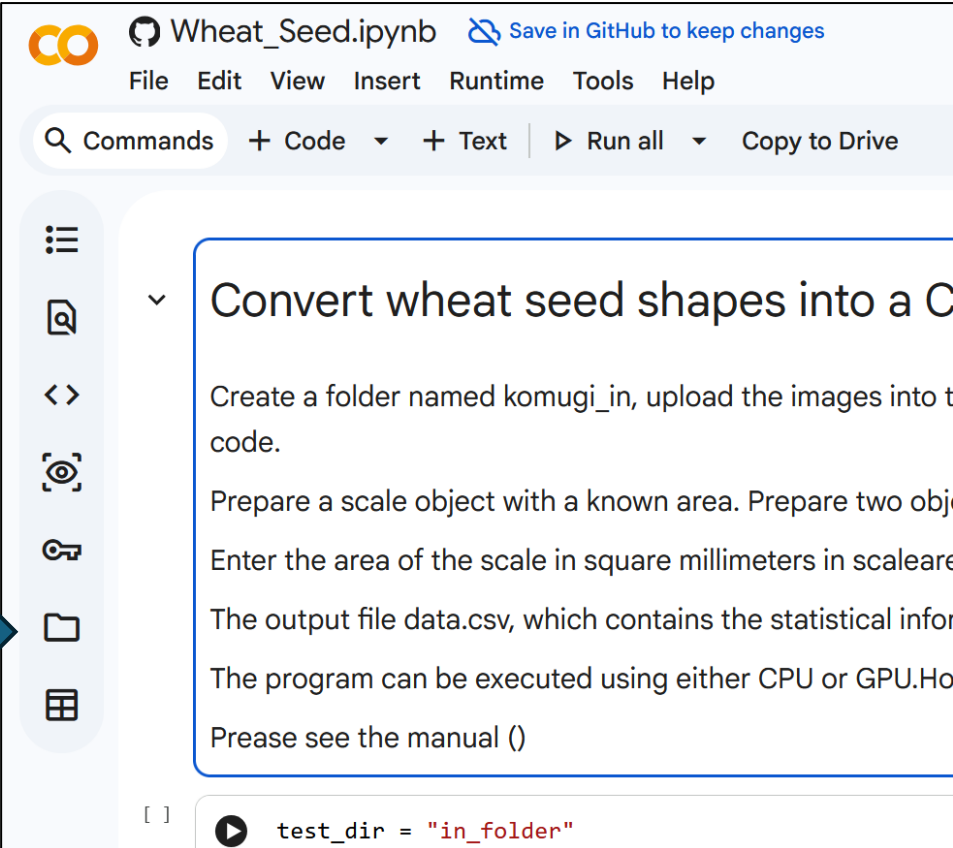
Place the seeds on a petri dish or similar dish and photograph them at the highest possible magnification. Even if the magnification varies between samples, this is corrected by the scale.



④ Run the script on google drive

Please refer to the sample data chapter and **configure it for the T4 GPU**. If the number of images is large, you may exhaust Google Colab's free quota. In that case, either purchase Compute units or download the script to your local environment and run it in Jupyter Lab. With 100 Compute units, you can analyze thousands to tens of thousands of images; upload speed will be the rate-limiting factor.

1



Wheat_Seed.ipynb Save in GitHub to keep changes

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Copy to Drive

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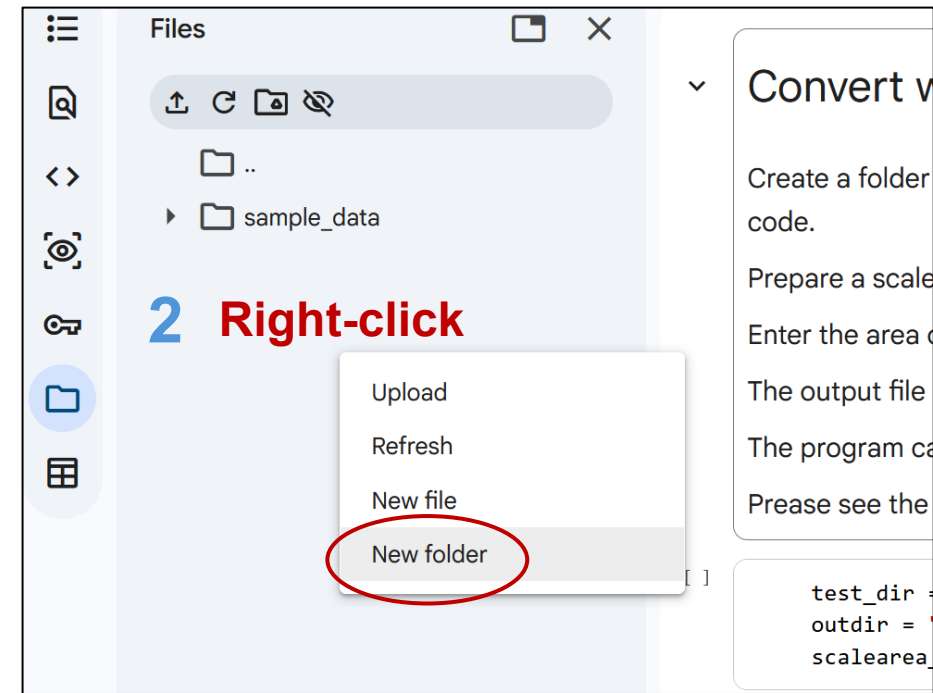
The output file data.csv, which contains the statistical infor

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Prease see the manual ()

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2 Right-click



Files

Upload Refresh New file New folder

Convert v

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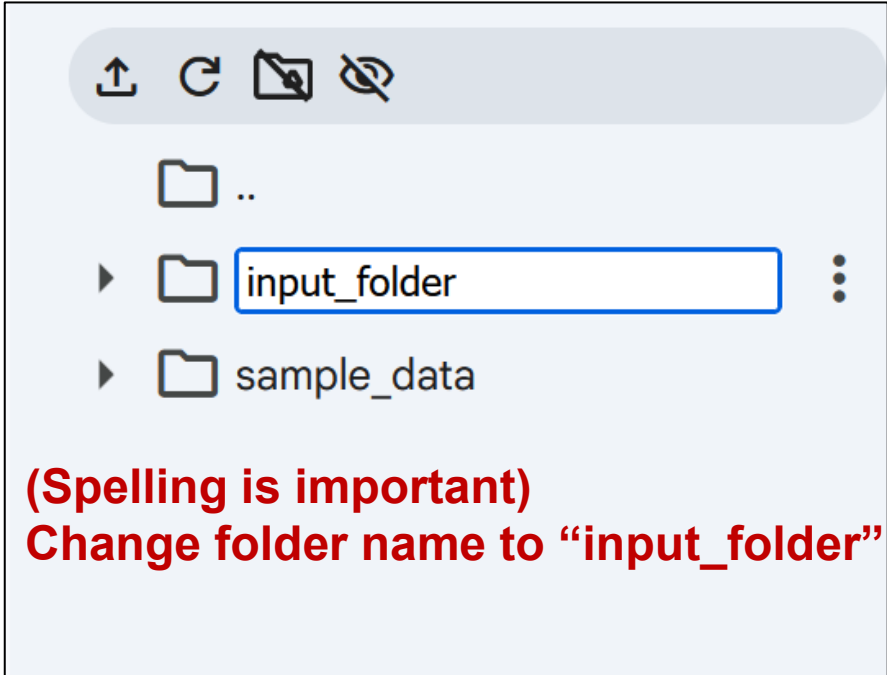
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The program ca

Prease see the

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test_dir =  
outdir =  
scalearea
```

3 (Spelling is important)
Change folder name to "input_folder"

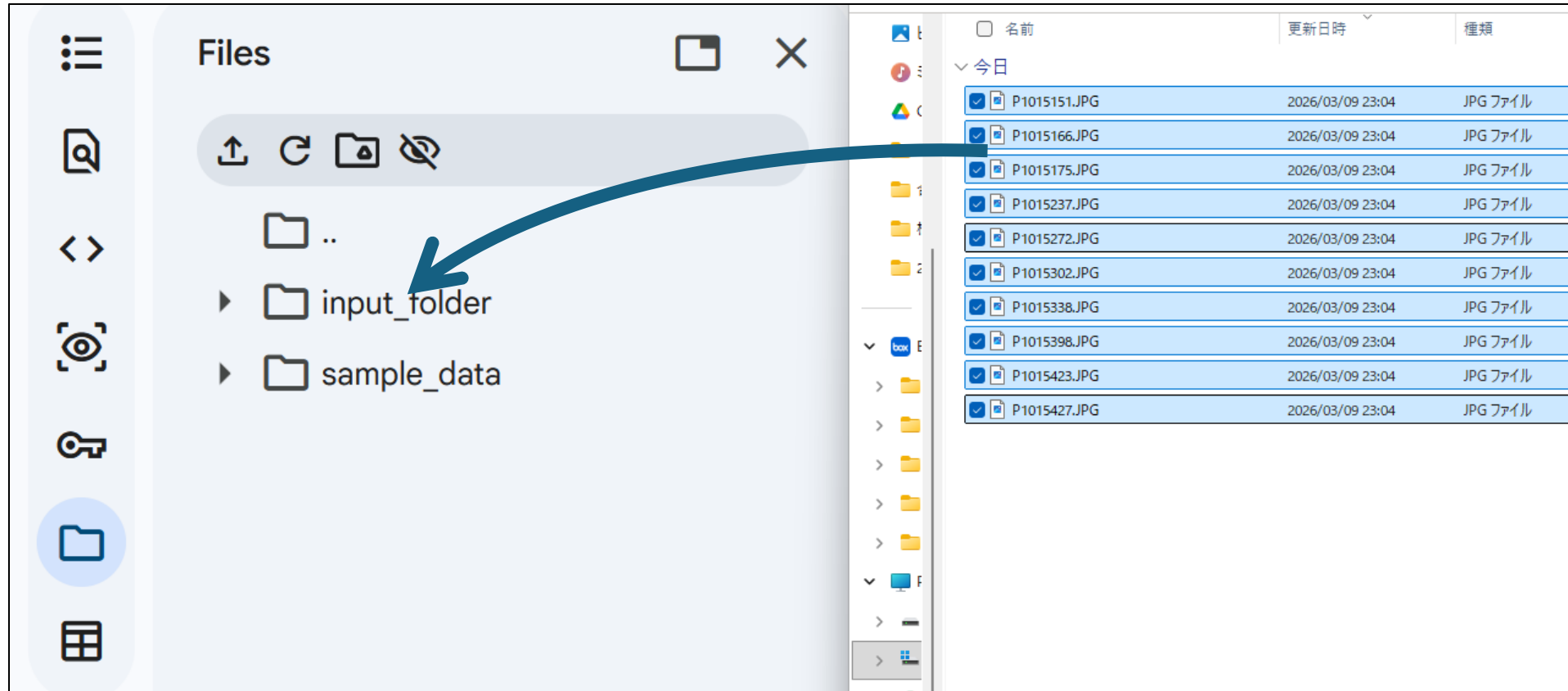


Files

input_folder sample_data

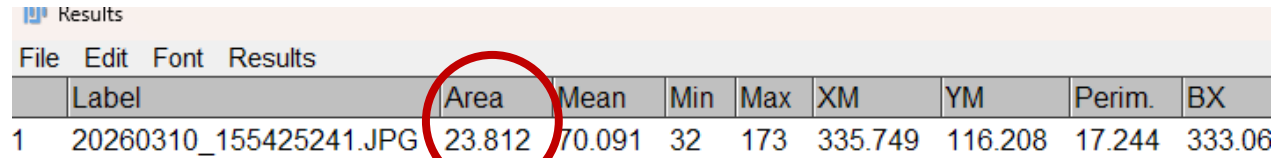
④ Run the script on google drive

4. Drag and drop pictures into “input_folder”



Uploading a large number of images takes considerable time,
So we recommend uploading them in batches of around 100 images.

④ Run the script on google drive



	Label	Area	Mean	Min	Max	XM	YM	Perim.	BX
1	20260310_155425241.JPG	23.812	70.091	32	173	335.749	116.208	17.244	333.06

]

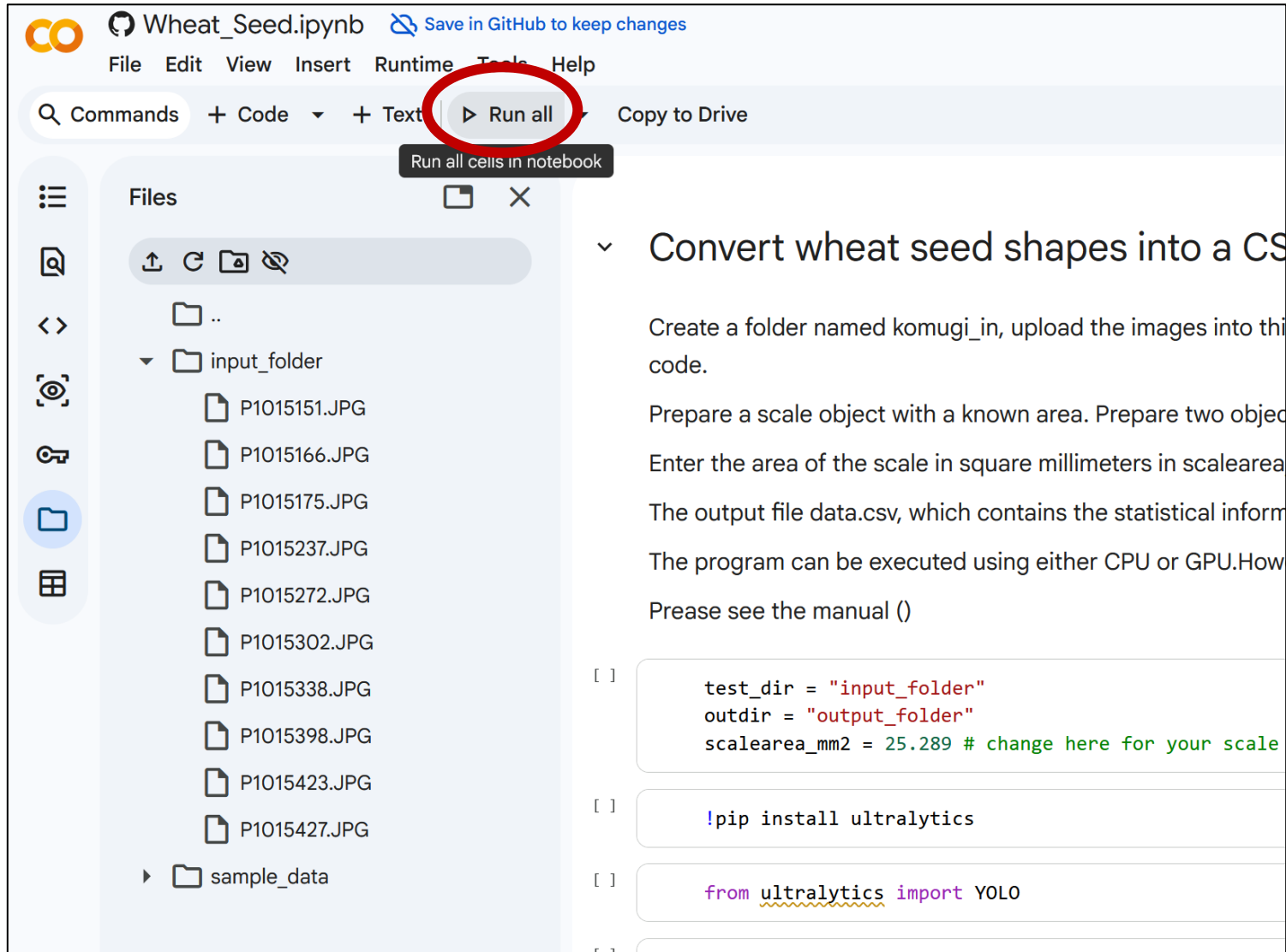
```
▶ test_dir = "input_folder"  
  outdir = "output_folder"  
  scalearea_mm2 = 23.812 # change here for your scale
```

5. Change here to scale size

④ Run the script on google drive

6. ▶ Run all

Please wait few minutes.



The screenshot shows a Jupyter Notebook interface for a file named 'Wheat_Seed.ipynb'. The top bar includes a search bar, a 'Commands' dropdown, and a 'Run all' button, which is circled in red. A tooltip for the 'Run all' button reads 'Run all cells in notebook'. The left sidebar displays a file explorer with a folder named 'input_folder' containing several JPEG files (P1015151.JPG to P1015427.JPG) and a 'sample_data' folder. The main area shows the notebook content, which includes a title 'Convert wheat seed shapes into a CS', a description of the task, and code cells. The first code cell contains the following Python code:

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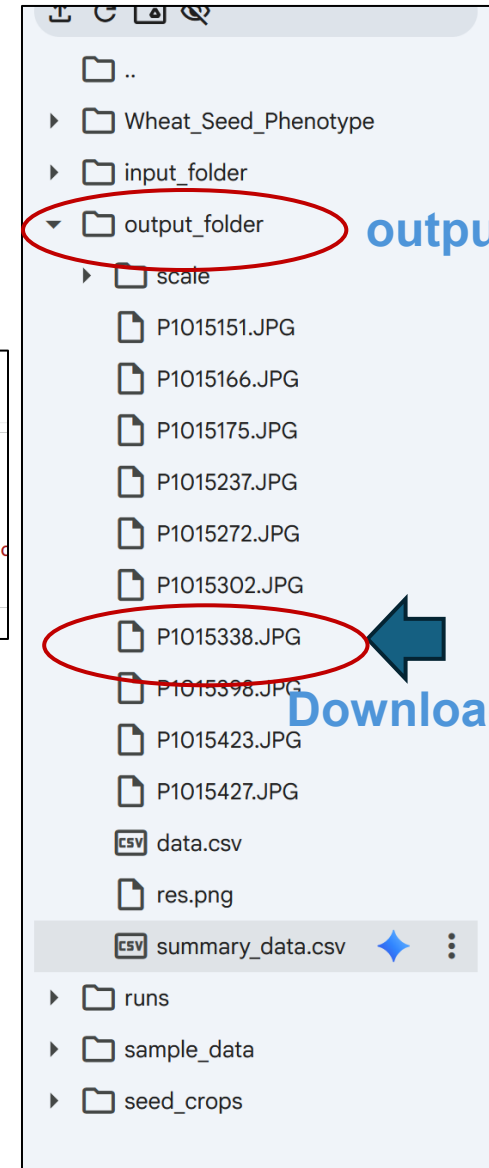
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④ Run the script on google drive

7. Scroll down and wait for the cell for
“Making summarized data” to complete.

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✓ [8] 0s Making summarized data  
import pandas as pd  
  
df = pd.read_csv(outdir + "/data.csv")  
result = df.groupby("File_Name")[["length_mm", "width_mm", "Area_mm2"]].agg(["mean", "std", "median", "count"])  
result.to_csv(outdir + "/summary_data.csv")
```

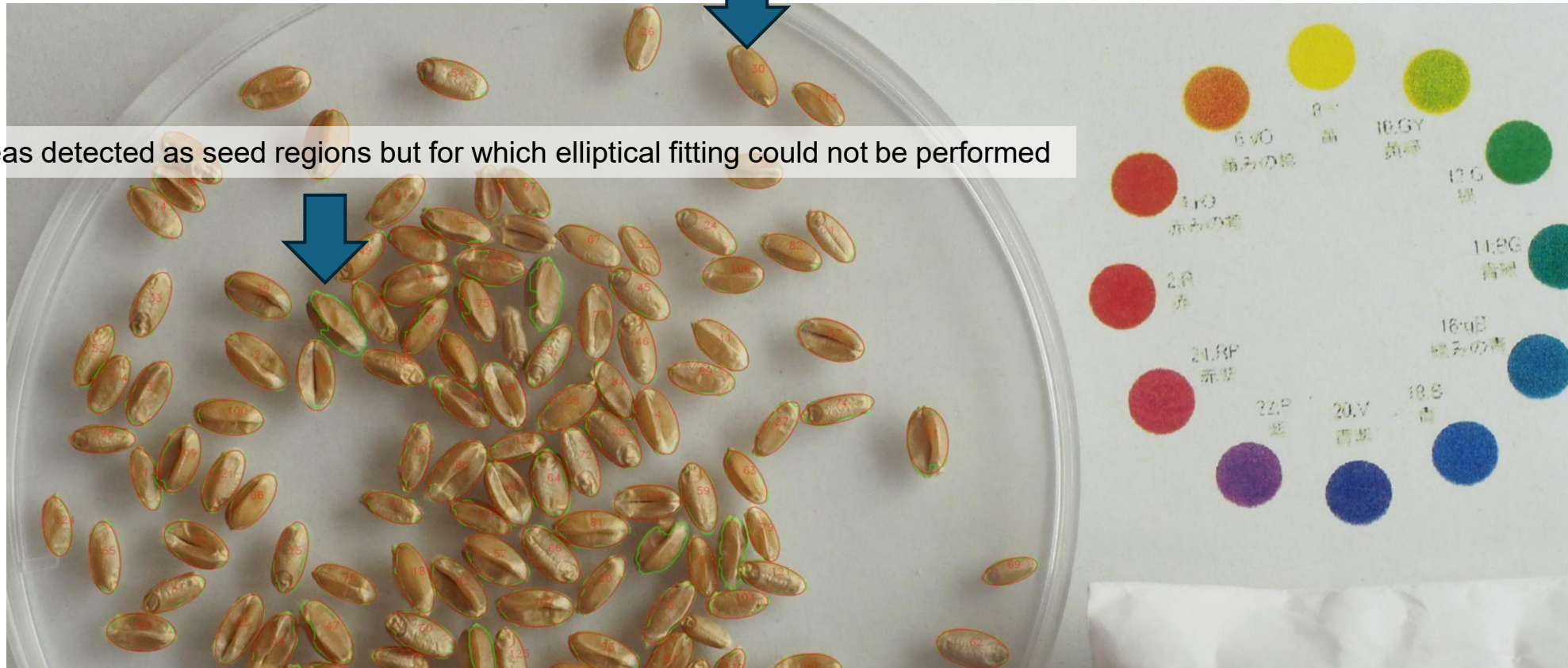


output_folder

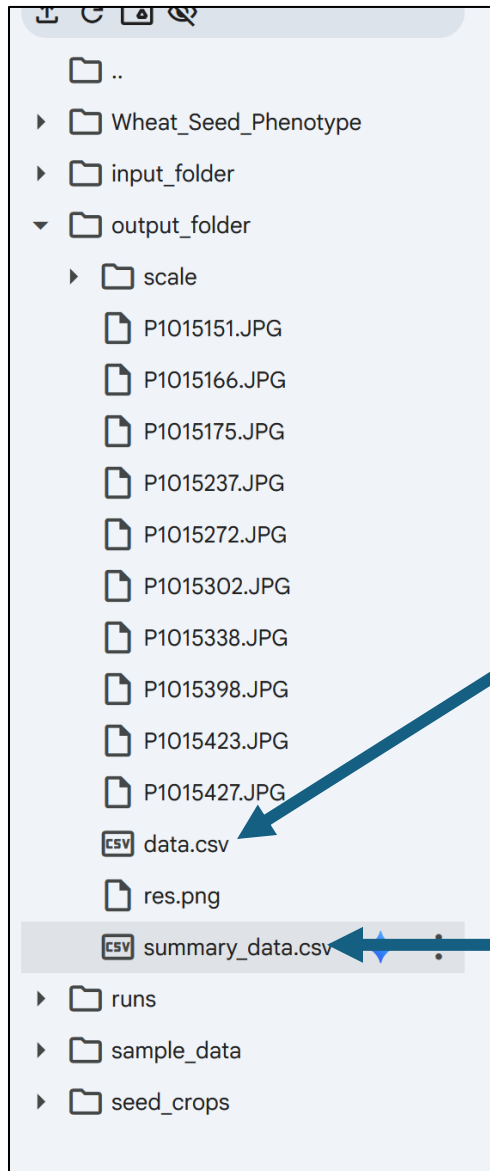
Download picture from output_folder

④ data check and analyze

Regions detected as seed regions where elliptical fitting was successful



④ data check and analyze



The Objective ID corresponds to the numerical value written within the image.

File_Name	Object_ID	Center_X	Center_Y	Major_Axi	Minor_Axi	Angle	Ellipse_Are	scale_px	px/mm	length_mm	width_mm	Ellipticity	Area_mm2
P1015302	1	1940.738	1001.328	153.1964	77.22924	128.3857	9292.234	16945.75	25.88598	5.918122	2.983439	0.504119	13.86727
P1015302	2	1871.732	1633.051	151.3884	78.98499	155.3967	9391.326	16945.75	25.88598	5.848276	3.051266	0.521738	14.01515
P1015302	3	741.5072	1371.947	147.9797	70.9954	142.1135	8251.298	16945.75	25.88598	5.716597	2.74262	0.479764	12.31383
P1015302	4	1650.505	1524.151	152.1355	82.4005	163.5266	9845.781	16945.75	25.88598	5.877138	3.18321	0.541626	14.69336
P1015302	5	1731.345	988.1266	168.6365	93.16708	52.97364	12339.68	16945.75	25.88598	6.514588	3.599133	0.552473	18.41513
P1015302	6	798.6462	895.4789	150.9453	80.94118	52.41896	9595.75	16945.75	25.88598	5.83116	3.126835	0.536229	14.32022
P1015302	7	1058.788	1132.315	143.2501	80.79838	136.7077	9090.495	16945.75	25.88598	5.533889	3.121318	0.564037	13.56621
P1015302	8	918.8369	1016.898	154.2676	93.95677	40.91505	11383.95	16945.75	25.88598	5.959506	3.62964	0.60905	16.98884

This data contains individual seeds data

Values in millimeters are here

This data contains aggregated data for each image.